

# BMJ Open Incidence, risk factors and pregnancy outcomes of gestational diabetes mellitus in Ibadan, Southwest Nigeria: a prospective cohort study

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## ABSTRACT

**Objective** Gestational diabetes mellitus (GDM) is an emerging public health concern in low and middle-income countries, including Nigeria, because of the associated pregnancy complications, increased healthcare costs and long-term health sequelae among women of reproductive age and their offspring. We determined the cumulative incidence, risk factors and pregnancy outcomes of GDM in Ibadan, Nigeria.

**Design** Prospective cohort study.

**Setting** Ibadan, Southwest Nigeria.

**Participants** 721 pregnant women from the Ibadan Pregnancy Cohort Study participated in the one-step 75 g-oral glucose tolerance test at 24–28 weeks' gestation.

**Outcomes** The primary outcome of the study is the cumulative incidence of GDM. GDM was diagnosed according to the International Association of Diabetes and Pregnancy Study Groups criteria. Secondary outcomes were pregnancy outcomes, which included modes of delivery (CS, spontaneous vaginal delivery), macrosomia (birth weight  $\geq 4.0$  kg), gestational age at delivery and birth asphyxia. The risk factors (exposures) examined included sociodemographic, obstetric, clinical, behavioural and lifestyle factors. Bivariate and multivariate Log-binomial regression models were used to identify the independent risk factors of GDM (adjusted for maternal age  $\geq 35$  years, income, maternal body mass index, history of pregnancy loss and congenital anomaly) and the associated pregnancy outcomes of GDM (adjusted for maternal age, income and maternal body mass index). Adjusted relative risk (aRR) and 95% CI, used to assess the strength of associations, were reported.

**Results** The cumulative incidence of GDM was 20.7%, 95% CI (17.9% to 23.9%). The mean time for the diagnosis of GDM is  $25.4 \pm 1.42$  weeks of gestation. After adjusting for other variables, maternal age  $\geq 35$  years (aRR: 1.48, 95% CI (1.07 to 1.97)  $p=0.016$ ), maternal obesity (aRR: 1.85, 95% CI (1.26 to 2.30)  $p=0.002$ ) and a previous history of congenital anomaly (aRR: 2.83, 95% CI (1.97 to 4.07)  $p<0.001$ ) were significantly associated with GDM risk. Women with GDM had a higher risk for elective CS: (RR 1.57 (95% CI: (1.04 to 2.36)  $p=0.032$ ), the association was insignificant after adjustment for other variables (aRR 1.32 (95% CI: (0.86 to 2.03)  $p=0.199$ ).

## STRENGTHS AND LIMITATIONS OF THE STUDY

- ⇒ Provision of gestational diabetes mellitus (GDM) data on cumulative incidence, risk factors and outcomes in low- and middle-income countries.
- ⇒ The use of the oral glucose tolerance test (OGTT) to diagnose GDM (one-step 75 g OGTT at 24–28 weeks of gestation, by the International Association of Diabetes and Pregnancy Study Groups criteria, which the WHO has adopted).
- ⇒ Our findings may be relevant to pregnant women who obtain care from comprehensive obstetric care in urban settings.
- ⇒ Our study may not be generalisable to pregnant women in other facilities (primary health care settings) or countries.
- ⇒ The biases include a potentially biased population, losses to follow-up and missing data from some pregnancy outcomes

**Conclusion** The cumulative incidence of GDM is high among pregnant women in Ibadan. Maternal age  $\geq 35$  years, maternal obesity and a history of congenital anomaly were significant independent risk factors for GDM. These factors should be targeted for public health interventions, including lifestyle modification among pregnant women with obesity and early screening and diagnosis of GDM.

## INTRODUCTION

Gestational diabetes mellitus (GDM) is defined as any degree of glucose intolerance with onset or first recognition during pregnancy.<sup>1–4</sup> It is the most common medical pregnancy complication.<sup>5 6</sup> In a healthy pregnancy, there are insulin-mediated physiologic adaptations of glucose, lipids and protein metabolism to meet the metabolic demands of pregnancy.<sup>7</sup> These adaptations are characterised by an early insulin sensitivity to substantial insulin resistance caused by placental hormones in the later half of

pregnancy till term.<sup>8 9</sup> However, despite the considerable insulin resistance, the blood glucose levels remain relatively constant because of a compensatory increase in the production of insulin by healthy beta-pancreatic cells. However, in GDM, the pancreatic beta cells' function is impaired, leading to the production of suboptimal insulin levels and resulting in hyperglycaemia.<sup>10–12</sup>

GDM has emerged as a global obstetric and public health challenge because of its rising prevalence as well as its adverse short-term and long-term consequences on the mother and her offspring.<sup>13</sup> Even though GDM is a global maternal health concern, there is a disproportionate rise in low- and middle-income countries (LMICs).<sup>14 15</sup> Globally, the prevalence of GDM is estimated to be 14% as reported by the International Diabetes Federation<sup>3</sup> according to WHO 2013/International Association of Diabetes and Pregnancy Study Groups (IADPSG) criteria.<sup>5 13</sup> However, there are variations across settings due to differences in screening and diagnostic criteria and underlying population characteristics such as socioeconomic status, lifestyle factors, access to health services, including social determinants such as poverty, education and income.<sup>16 17</sup> The International Diabetes Federation Atlas 11th edition (2025) highlights the rising prevalence of GDM, affecting one in six livebirths worldwide, and 23.3million livebirths (15.6%) were exposed to some form of hyperglycaemia, mainly in the LMICs.<sup>5 18</sup> The prevalence across continents has been reported as South Asia (24.9%), North America and the Caribbean (17.9%), Europe (13.9%), South and Central America (12.5%), sub-Saharan Africa (SSA) (10.2%) and the Middle East and North Africa (10.2%).<sup>18</sup> Within the SSA, the prevalence in countries include Ethiopia (14.9%), Nigeria (12.4%), South Africa (11.5%), Tanzania (8.1%), and Kenya (4.0%).<sup>19</sup> The high burden of GDM is corroborated by a meta-analysis on GDM in Nigeria (2021) that reported a pooled prevalence of 11%.<sup>20</sup>

The escalating burden of GDM in LMIC, such as Nigeria, is driven by an ongoing epidemiological and nutritional transition that has led to an increase in the occurrence of obesity and diabetes in the general population and women of reproductive age.<sup>21 22</sup> Generally, the risk factors of GDM have been classified into modifiable and non-modifiable factors.<sup>23</sup> The non-modifiable factors include advanced maternal age, family history of diabetes, multiparity, histories of previous gestational diabetes, stillbirth, congenital anomaly, macrosomia and polycystic ovary syndrome. The modifiable factors are maternal overweight and obesity, excessive gestational weight gain, unhealthy diet, physical inactivity and passive smoking, a rise of undetected dysglycaemia among women of reproductive age.<sup>17 24–26</sup> Notably, excess adiposity and body-weight have been the strongest and most consistent risk factors.<sup>23</sup>

The significance of GDM arises from the serious consequences during pregnancy in the mother and the offspring, as well as programming both for future chronic diseases.<sup>27</sup> The short-term complications of GDM

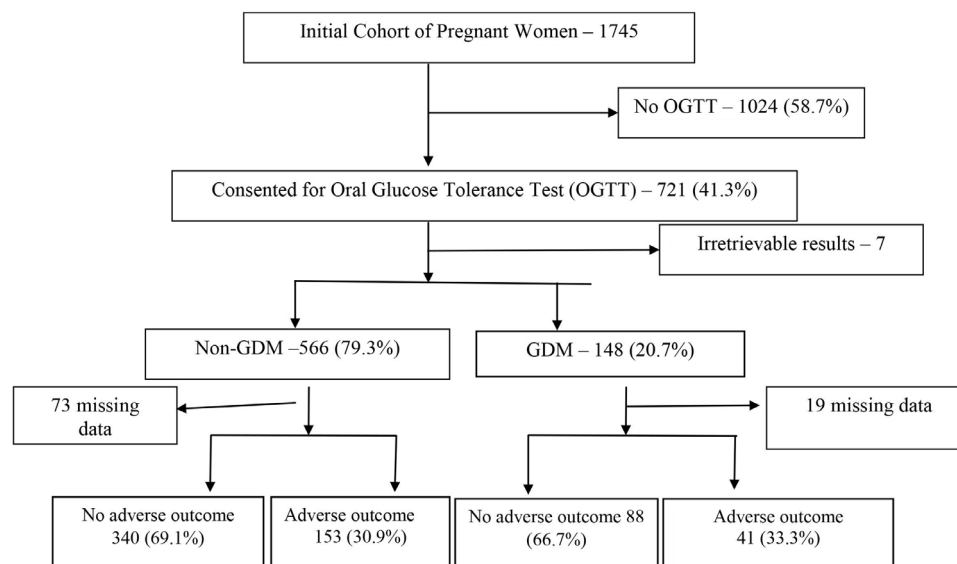
include obstructed labour, increased CS rates, shoulder dystocia, birth trauma, nerve palsy, the increased risk of congenital anomalies and stillbirth in severe cases,<sup>28</sup> increased operational deliveries, increased admission rate into the neonatal intensive care unit, infant respiratory distress syndrome and neonatal jaundice.<sup>29 30</sup> The long-term maternal complications include an increased risk of type two diabetes mellitus, metabolic syndrome and cardiovascular diseases, such as<sup>2</sup> myocardial infarction and stroke.<sup>31–33</sup> Several studies have also shown that GDM increases the risk of overweight, obesity, metabolic syndrome and type 2 diabetes in the offspring in later life.<sup>34 35</sup>

The oral glucose tolerance test (OGTT) is used to diagnose GDM at glucose values elevated above the prespecified values,<sup>36</sup> and it is performed using either one-step or two-step approaches.<sup>17</sup> The IADPSG criteria, adopted by the American Diabetes Association, WHO and the International Federation of Gynaecology and Obstetrics, are the most commonly used threshold.<sup>37</sup> Notably, while universal screening is recommended and practised in high-income countries (HICs), a selective screening strategy based on the presence of risk factors is adopted in LMICs due to inadequate resources, including a trained workforce, limited access to screening and increased workload from a number of cases.<sup>38</sup> Already, Nigeria has the highest maternal mortality ratio in SSA (574 per 100 000 live births) and a neonatal mortality rate of 34.9 per 1000 live births.<sup>39</sup> A systematic review and meta-analysis of GDM in Nigeria reported a GDM prevalence of 0.5%–38% and a pooled prevalence of 11.4%, highest in the Federal Capital Territory at 16%.<sup>20</sup> GDM can further worsen maternal and neonatal morbidity and mortality and increase the future risk of non-communicable diseases in Nigeria. Additionally, neglecting the GDM in Nigeria would place a considerable strain on the health sector/system due to the resources, personnel, medicines and technologies required to manage hyperglycaemia and its complications.<sup>40</sup> Therefore, in this study, we sought to determine the incidence, risk factors and associated pregnancy outcomes of GDM in Ibadan, Southwest Nigeria, using a prospective cohort study among pregnant women in Ibadan—the Ibadan Pregnancy Cohort Study (IbPCS)<sup>41</sup>

## METHODS

### Study design and setting

The IbPCS is a prospective cohort study of pregnant women that investigated the associations between maternal obesity and lifestyle factors on glycaemic control, gestational weight gain and pregnancy outcomes. This study was conducted among pregnant women obtaining care from any of four comprehensive obstetric health facilities in Ibadan, Southwest Nigeria, from April 2018 to September 2019. The details of the study (IbPCS) have been described elsewhere.<sup>41</sup> The study was conducted in Ibadan, Oyo State, Southwestern Nigeria,



**Figure 1** The flowchart of participants' uptake of OGTT, GDM diagnosis and pregnancy outcomes. GDM, gestational diabetes mellitus.

which is the third-most populous city in Nigeria, with a population exceeding 5 million people.<sup>39</sup> These facilities and the average delivery per year were as follows: University College Hospital (1500), Adeoyo Maternity Teaching Hospital (2500), Jericho Specialist Hospital (1200) and Saint Mary Catholic Hospital, Oluoyo (800). The number of pregnant women recruited from each health facility was proportionately allocated according to the facilities' yearly delivery population: University College Hospital (442), Adeoyo Maternity Teaching Hospital (757), Saint Mary Catholic Hospital, Oluoyo (99) and Jericho Specialist Hospital (447). The women were recruited at their first antenatal visit during early gestation ( $\leq 20$  weeks' gestation) and followed up until delivery. The women's baseline characteristics, including sociodemographic, lifestyle, obstetric and medical histories, were ascertained during enrolment. The flow chart for the Ibadan Pregnancy Cohort regarding the uptake of OGTT, diagnosis of GDM and associated outcomes is shown in figure 1.

### Source population

All pregnant women who obtain antenatal and delivery care at federal and state-owned general and teaching hospitals in South-West, Nigeria.

### Study population

The study population comprised all pregnant women who registered (booked) for antenatal care at the selected health facilities within the  $\leq 20$  weeks of the index pregnancy.

### Inclusion criteria

Pregnant women who were within the first 20 weeks of gestation, aged  $\geq 18$  years,

### Exclusion criteria

Pregnant women with medical disorders (pre-existing diabetes mellitus, severe hypertension, renal disease).

### Sample size determination and sampling techniques

The sample size for the IbPCS was estimated using the formula for comparing two proportions already reported in the protocol.<sup>41</sup> Overall, 1745 pregnant women who met the eligibility criteria were enrolled at their first antenatal booking visit. These pregnant women, who were recruited into the study from their first A(ANC) visit ( $\leq 20$  weeks' gestation), were followed up until delivery. All participants were invited to an OGTT between 24 weeks and 28 weeks, but only 41.3% participated.

### Variables of study

#### Dependent variables

1. Primary outcome: GDM, defined as surpassing one or more of the thresholds—fasting plasma glucose (FPG)  $\geq 5.1$  mmol/L; 1-hour plasma glucose  $\geq 10.0$  mmol/L, 2-hour plasma glucose  $\geq 8.5$  mmol/L,<sup>42</sup> according to the IADPSG recommendations.
2. Secondary outcomes: pregnancy outcomes—(1) *caesarean section (CS)*; (2) *spontaneous vertex delivery*; (3) *induction of labour*; (4) *macrosomia*: birthweight  $\geq 4.0$  kg; (5) *low birth weight*: birthweight  $\leq 2.50$  kg (6) *birth asphyxia at 1 min*: APGAR score  $\leq 7$  at 1 min; (7) *preterm delivery*: delivery  $< 37$  completed weeks of gestation; (8) *postpartum haemorrhage*: blood loss post vaginal delivery  $\geq 500$  mL and  $\geq 1000$  mL post CS.

#### Independent variables (exposures)

(1) Sociodemographic characteristics: maternal age, marital status, level of education, employment status, monthly income, religion.



**Table 1** Baseline characteristics of the study participants, Ibadan, Nigeria

| Characteristics          | N 714 (frequency) | Percentage (n/N) |
|--------------------------|-------------------|------------------|
| Age group                |                   |                  |
| < 35 years               | 555               | 77.7             |
| ≥35 years                | 159               | 22.3             |
| Mean age (years)         | 30.0 (±5.2)       |                  |
| Parity                   |                   |                  |
| Nulliparous              | 315               | 44.2             |
| 1–3                      | 351               | 49.3             |
| ≥4                       | 46                | 4.5              |
| Marital status           |                   |                  |
| Single                   | 46                | 6.4              |
| Married                  | 668               | 93.6             |
| Level of education       |                   |                  |
| At least primary         | 18                | 2.5              |
| Secondary                | 206               | 28.9             |
| Tertiary                 | 488               | 69.6             |
| Employment status        |                   |                  |
| Employed                 | 632               | 88.5             |
| Unemployed               | 82                | 11.5             |
| Religion                 |                   |                  |
| Christianity             | 420               | 58.3             |
| Islam                    | 288               | 40.7             |
| Income per month (naira) |                   |                  |
| <20 000                  | 232               | 37.3             |
| 20 000–99 999            | 344               | 55.2             |
| ≥1 00 000                | 47                | 7.5              |
| BMI                      |                   |                  |
| Underweight              | 30                | 4.3              |
| Normal weight            | 349               | 50.4             |
| Overweight               | 193               | 27.8             |
| Obese                    | 232               | 17.5             |

BMI, body mass index.

(2) Obstetric and clinical characteristics: parity, histories of pregnancy losses (miscarriage/stillbirth), congenital anomalies, macrosomia, GDM, family history of DM and depression.

(3) Behavioural and lifestyle variables: maternal body mass index, alcohol consumption, tobacco exposure, physical activity, fast food consumption.

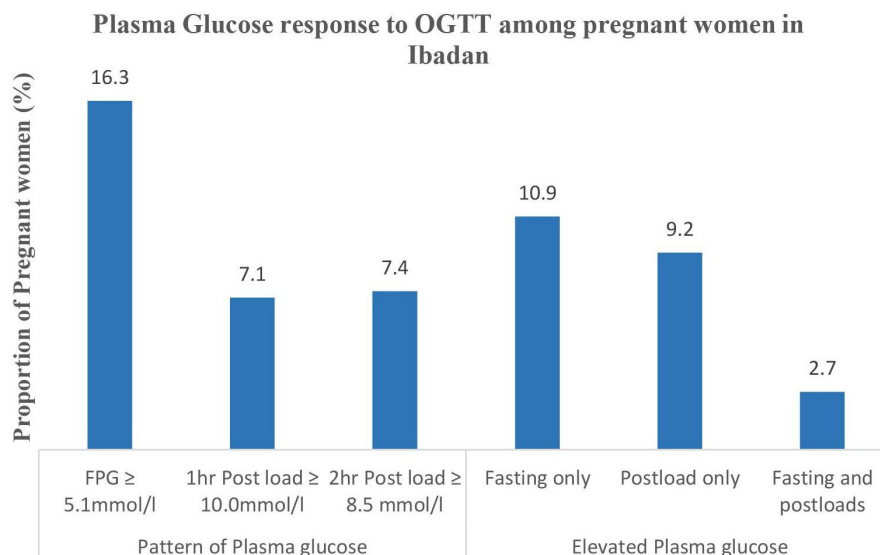
### Data collection procedures

Trained research nurses coordinated the data collection process. Data were collected using pretested, interviewer-administered questionnaires and structured proformas at booking, the third trimester and delivery. The women's baseline characteristics, including sociodemographic,

maternal lifestyle characteristics, obstetric and medical histories, were ascertained during enrolment through interviewer-administered questionnaires. The maternal anthropometric (weight and height) was measured by research nurses using standard procedures at recruitment, after which maternal weight was assessed at every antenatal care visit. We determined the body mass index (BMI) in kg/m<sup>2</sup> according to the WHO classification.<sup>43 44</sup> The physical activity level was assessed using the duration of moderate-intensity exercise in minutes, according to the guidelines of the Pregnancy Physical Activity Questionnaire.<sup>45</sup> We also captured the duration of television viewing and sleep duration (in hours). We measured the fast-food consumption as either frequent (weekly or more) or not frequent (monthly or less). Alcohol users are referred to as those who consumed alcohol during the index pregnancy. At the same time, tobacco exposure, including current smokers, exposure to secondhand smoke from a spouse and those exposed to smokeless tobacco (self-reports).<sup>46</sup> Antepartum depression was ascertained during the third trimester using the Edinburgh Postnatal Depression Scale (EPDS), with EPDS ≥12 for major depressive disorders.<sup>47</sup> The information on pregnancy outcomes was abstracted from the medical records.

### Assessment of GDM

The assessment of GDM was performed using a 75 g OGTT, done between 24 weeks and 28 weeks of gestation. At recruitment, participants were informed about their participation in an OGTT scheduled for 24–28 weeks. Study participants were reminded of the test a few days in advance and instructed to fast for at least 10–12 hours overnight. The procedure is conducted by the Research Nurse and the Laboratory Scientist, who performed the phlebotomy and sample preparation. The women arrived early on the morning of the test; their vital signs and weight were measured after they had rested for about 10 min. Afterwards, 4 mL of venous blood was withdrawn from each participant and dispensed into a fluoride oxalate specimen bottle for FPG estimation. Then 420 mL of commercial glucose preparation (equivalent to 75 g of glucose in 300 mL of water) was measured and given to participants to drink within a 5-minute duration. 4 mL of venous blood was also withdrawn at 1 hour and 2 hours post glucose load and dispensed into separate well-labelled fluoride oxalate specimen bottles. The withdrawn blood specimen was transported in a special container called Giostat containing ice-packs to the chemical pathology laboratory of the University College Hospital immediately after collection. Each specimen tube was centrifuged for 10 min at 3000 g within 30–60 min post collection, and the plasma was separated and dispensed into plain tubes for glucose analysis. The plasma was stored at –200°C and analysed for glucose within 3 months of collection. The OGTT procedure was conducted at each facility every week by the research nurse and laboratory scientist. The plasma glucose was assayed using the glucose oxidase



**Figure 2** Plasma glucose response to OGTT among pregnant women. FPG, fasting plasma glucose.

method on an automated chemistry analyser, Landwind C100 plus.

### Data quality control

The study instruments were pretested before the study commenced, and trained research nurses and assistants conducted data collection. We undertook regular checks to ensure completeness and consistency. We strictly complied with the manufacturers' instructions and standard operating procedures.

### Data processing and analysis

Data were entered using SPSS V.20 (IBM, New York, USA, 2013), cleaned and checked for missing values and exported to STATA V.13 (Stata Corp LLC, College Station, Texas, USA, 2015) for statistical analysis. Descriptive statistics were applied using means and SD to summarise continuous variables and proportions and percentages for categorical variables. Bivariate log-binomial regression models were employed to assess the relationship between the independent risk factors, that is, sociodemographic, clinical, obstetric, behavioural and lifestyle factors and GDM; unadjusted relative risk (uRR) estimates, uRR, 95% CIs and p values were reported. Factors that were significant at a 10% significance level—maternal age  $\geq 35$  years, income, maternal BMI, history of pregnancy losses and history of congenital anomaly—were entered into the multivariable log binomial regression model. Adjusted relative risks (aRRs) were used to estimate the strength of association between the independent risk factors, after controlling for possible confounding factors. Furthermore, we evaluated the association between GDM and pregnancy outcomes also using bivariate log-binomial regression models and adjusted for important confounders—maternal age, income and maternal BMI using multivariable log-binomial regression models. uRR and aRR, 95% CIs and p values were reported, and variables with a p value  $<0.05\%$  were regarded as significant.

### Patient and public involvement

Patients and the public were not involved in the planning, design or interpretation of the analysed data.

## RESULTS

### Characteristics of study participants

The baseline characteristics of the study participants are shown in table 1. The mean age was  $30.0 \pm 5.2$  years. The majority of participants were married 668 (93.6%), employed 632 (88.5%) and had a tertiary-level education 488 (68.5%). The distribution of monthly maternal income was  $<20\,000$  naira, 232 (37.3%);  $\#20\,000$ – $\#99\,999$  naira, 344 (55.2%);  $\geq 100\,000$  naira, 47 (7.5%). Women who were overweight or obese were 193 (27.8%) and 132 (17.5%), respectively. The plasma glucose response to OGTT among pregnant women is shown in figure 2. The proportion of study participants with plasma glucose levels exceeding thresholds is as follows: FPG  $\geq 5.1$  mmol/L—16.3%; 1-hour plasma glucose  $\geq 10.0$  mmol/L—7.1%; 2-hour plasma glucose  $\geq 8.5$  mmol/L—7.4%. However, only 10.9% and 9.7% had elevated FPG and post load glucose, respectively. Only 2.7% had elevated values over all the diagnostic thresholds.

### Cumulative Incidence of GDM

Among the 714 pregnant women who had undergone OGTT, 148 were diagnosed with GDM. The mean gestational age for the diagnosis of GDM is  $25.4 \pm 1.42$  weeks of gestation. The cumulative incidence was 20.7% (95% CI: 17.9% to 23.7%) over 6 months.

### Sociodemographic characteristics of pregnant women with or without GDM

Sociodemographic characteristics of pregnant women with or without GDM are shown in table 2. The proportion of older women (aged  $\geq 35$  years) diagnosed with GDM was compared with that of those without GDM

**Table 2** Sociodemographic characteristics of pregnant women by their GDM status in Ibadan, Southwest Nigeria

|                             | Total      | Gestational diabetes mellitus |                   | Unadjusted relative risk<br>95% CI | P value       | Adjusted relative risk<br>95% CI | P value       |
|-----------------------------|------------|-------------------------------|-------------------|------------------------------------|---------------|----------------------------------|---------------|
|                             |            | Present N (%)                 | Absent N (%)      |                                    |               |                                  |               |
| <b>Cumulative incidence</b> | <b>714</b> | <b>148 (20.7)</b>             | <b>566 (79.3)</b> |                                    |               |                                  |               |
| Age group                   |            |                               |                   |                                    |               |                                  |               |
| < 35 years                  | 555 (77.7) | 103 (69.6)                    | 452 (79.9)        | 1                                  |               |                                  |               |
| ≥ 35 years                  | 159 (22.3) | 45 (30.4)                     | 114 (20.1)        | 1.53 (1.13 to 2.06)                | <b>0.006*</b> | 1.48 (1.07 to 1.97)              | <b>0.016*</b> |
| Mean age                    | 30.0±5.4   | 30.9±6.3                      | 29.8±5.2          |                                    | <b>0.028*</b> |                                  |               |
| Marital status              |            |                               |                   |                                    |               |                                  |               |
| Single                      | 46 (6.4)   | 5 (3.4)                       | 41 (7.2)          | 1                                  |               |                                  |               |
| Married                     | 668 (93.6) | 143 (96.6)                    | 525 (92.8)        | 1.97 (0.85 to 4.56)                | 0.114         |                                  |               |
| Maternal education          |            |                               |                   |                                    |               |                                  |               |
| Primary                     | 18 (2.5)   | 4 (2.7)                       | 14 (2.5)          | 1                                  |               |                                  |               |
| Secondary                   | 206 (28.9) | 42 (28.4)                     | 164 (29.1)        | 0.98 (0.57 to 2.27)                | 0.852         |                                  |               |
| Tertiary                    | 488 (68.5) | 102 (68.9)                    | 386 (68.4)        | 0.97 (0.39 to 2.27)                | 0.892         |                                  |               |
| Employment status           |            |                               |                   |                                    |               |                                  |               |
| Unemployed                  | 82 (11.5)  | 19 (12.8)                     | 63 (11.3)         | 1                                  |               |                                  |               |
| Employed                    | 632 (88.5) | 129 (87.2)                    | 503 (88.9)        | 0.88 (0.58 to 1.34)                | 0.557         |                                  |               |
| Religion                    |            |                               |                   |                                    |               |                                  |               |
| Christianity                | 420 (59.3) | 87 (59.6)                     | 333 (59.3)        | 1                                  |               |                                  |               |
| Islam                       | 288 (40.7) | 59 (40.4)                     | 229 (40.7)        | 0.99 (0.74 to 1.33)                | 0.941         |                                  |               |
| Income (naira)              |            |                               |                   |                                    |               |                                  |               |
| <20 000                     | 232 (37.2) | 37 (29.8)                     | 195 (39.1)        | 1.                                 |               | 1                                |               |
| 20 000–99 999               | 344 (55.3) | 44 (59.7)                     | 270 (54.1)        | 1.34 (0.94 to 1.92)                | 0.101         | 1.24 (0.86 to 1.77)              | 0.243         |
| ≥1 00 000                   | 47 (7.5)   | 13 (10.5)                     | 34 (6.8)          | 1.73 (1.01 to 3.00)                | <b>0.049*</b> | 1.26 (0.72 to 2.22)              | 0.424         |

Bold values indicate a statistically significant association between exposures and GDM

(30.4% vs 20.1%). Similarly, higher income-earning women (≥ 100 000 naira) were more prevalent among women with GDM compared with those without GDM (10.5% vs 6.8%). However, the distribution of marital status, maternal education and employment status was similar among women with and without GDM.

#### Clinical and obstetric characteristics of pregnant women with or without GDM

Clinical and obstetric characteristics of pregnant women with or without GDM are displayed in [table 3](#): multiparous women among women with and without GDM (4.8% vs 6.9%), history of pregnancy losses (25% vs 29.7%), history of congenital anomaly (1.4% vs 0.4%), history of macrosomia (8.1% vs 8.0%) and a family history of DM (8.8% vs 9.7%) among the respondents. The occurrence of depression was slightly higher in women with than those without GDM (18.1% vs 16.3%).

#### Behavioural and lifestyle characteristics of pregnant women with or without GDM

Behavioural and lifestyle characteristics of pregnant women with or without GDM are shown in [table 4](#). Alcohol consumption (16.9% vs 13.4%), tobacco exposure (5.4%

vs 3.4%), frequent fast-food consumption (17.1% vs 16.0%), maternal BMI among women with and without was as follows: underweight (17.1% vs 16.0%), normal weight (43.8% vs 52.1%), overweight (27.1% vs 28.1%) and obese (27.1% vs 14.9%). Women with depression (18.1% vs 16.3%). The mean duration of moderate intensity activity (25.3±3.5) minutes per week, time spent in television viewing (2.5±3.5) hours per day and sleeping (7.9±1.7) hours per night, but there were no significant differences between the two groups.

#### Risk factors of GDM

The risk factors of GDM among pregnant women in Ibadan are shown in [tables 2–4](#). After adjusting for maternal age ≥35 years, maternal income, maternal BMI, history of pregnancy losses and congenital anomaly, the adjusted log-binomial model showed that maternal age ≥35 years (aRR: 1.48, 95% CI: (1.07 to 1.97) p=0.016), maternal obesity (aRR: 1.85, 95% CI (1.26 to 2.30) p=0.002) and a history of congenital anomaly (aRR: 2.83, 95% CI (1.97 to 4.07) p<0.001) were significantly associated with GDM.

**Table 3** Obstetric and clinical characteristics of pregnant women by their GDM status in Ibadan, Southwest Nigeria

|                               | Total      | GDM               |                   | Unadjusted<br>Relative Risk<br>95% CI | P value | Adjusted Relative<br>Risk<br>95% CI | P value          |
|-------------------------------|------------|-------------------|-------------------|---------------------------------------|---------|-------------------------------------|------------------|
|                               |            | Present N (%)     | Absent N (%)      |                                       |         |                                     |                  |
| Cumulative Incidence          | <b>714</b> | <b>148 (20.7)</b> | <b>566 (70.3)</b> |                                       |         |                                     |                  |
| Parity                        |            |                   |                   |                                       |         |                                     |                  |
| Nullipara                     | 315 (44.2) | 63 (42.9)         | 252 (44.6)        | 1                                     |         |                                     |                  |
| 1–3                           | 351 (49.3) | 77 (52.4)         | 274 (48.5)        | 1.10 (0.82 to 1.47)                   | 0.541   |                                     |                  |
| ≥4                            | 49 (6.5)   | 7 (4.8)           | 39 (6.9)          | 0.76 (0.37 to 1.59)                   | 0.455   |                                     |                  |
| History of pregnancy losses*  |            |                   |                   |                                       |         |                                     |                  |
| No                            | 509 (71.3) | 111 (75.0)        | 398 (70.3)        | 1                                     |         | 1                                   |                  |
| Yes                           | 205 (28.7) | 37 (25.0)         | 168 (29.7)        | 0.83 (0.59 to 1.15)                   | 0.068   | 0.74 (0.54 to 1.01)                 | 0.063            |
| History of congenital anomaly |            |                   |                   |                                       |         |                                     |                  |
| No                            | 710 (99.4) | 146 (98.7)        | 564 (99.6)        | 1                                     |         | 1                                   |                  |
| Yes                           | 4 (0.6)    | 2 (1.4)           | 2 (0.4)           | 2.43 (0.09 to 6.54)                   | 0.079   | 2.83 (1.97 to 4.07)                 | <b>&lt;0.001</b> |
| History of macrosomia         |            |                   |                   |                                       |         |                                     |                  |
| No                            | 657 (92.0) | 136 (91.9)        | 521 (88.5)        | 1                                     |         |                                     |                  |
| Yes                           | 57 (8.0)   | 12 (8.1)          | 45 (8.0)          | 1.01 (0.60 to 1.71)                   | 0.950   |                                     |                  |
| History of GDM                |            |                   |                   |                                       |         |                                     |                  |
| Yes                           | 6 (1.2)    | 2 (2.0)           | 4 (1.1)           | 1                                     |         |                                     |                  |
| No                            | 477 (98.8) | 99 (98.0)         | 378 (98.9)        | 1.61 (0.57 to 5.21)                   | 0.417   |                                     |                  |
| Family history of DM          |            |                   |                   |                                       |         |                                     |                  |
| No                            | 646 (90.5) | 135 (91.2)        | 511 (90.3)        | 1                                     |         |                                     |                  |
| Yes                           | 68 (9.5)   | 13 (8.8)          | 55 (9.7)          | 0.91 (0.52 to 1.52)                   | 0.731   |                                     |                  |
| Depression                    |            |                   |                   |                                       |         |                                     |                  |
| No                            | 560 (83.3) | 113 (80.9)        | 447 (83.7)        |                                       |         |                                     |                  |
| Yes                           | 112 (16.7) | 25 (18.1)         | 87 (16.3)         | 1.11 (0.75 to 1.62)                   | 0.645   |                                     |                  |

Bold values indicate a statistically significant association between exposures and GDM

\*History of pregnancy losses: women with a previous history of miscarriage or prior history of stillbirth.

DM, diabetes mellitus; GDM, gestational DM.

### Associations between GDM and pregnancy outcomes

The associations between GDM and pregnancy outcome in Ibadan are shown in table 5. The cumulative incidence of CS (33.3% vs 30.9%), elective CS (20.8% vs 13.3%), macrosomia (7.2% vs 4.9%) and preterm delivery (14.3% vs 11.7%) was higher among women with GDM compared with women without GDM. Although the risk of an elective CS was significantly higher among women with GDM (uRR 1.57 (95% CI: (1.04 to 2.36); p value: 0.032) compared with women without GDM. After adjusting for maternal age ≥35 years, maternal income and maternal BMI, the adjusted log-binomial model showed that the risk of an elective CS was no longer significant (aRR 1.32 (95% CI: (0.86 to 2.03); p value: 0.316).

### DISCUSSION

In this study, we investigated the incidence, risk factors and associated pregnancy outcomes of GDM in Ibadan, Nigeria. We found a cumulative incidence of GDM of 20.7%, indicating that one in every five pregnant women

in our setting had GDM. Additionally, advanced maternal age, maternal obesity and a prior history of congenital anomaly were significant risk factors of GDM; maternal income and a history of pregnancy losses were insignificant after adjusting for confounders. Regarding adverse pregnancy outcomes, none had a significant association with GDM. Generally, the incidence of GDM is increasing in SSA, according to a recent meta-analysis that analysed the wide variations in the incidence of GDM.<sup>19</sup> The incidence of GDM was 20.7% in our study population, which was similar to that reported in Sudan (18.5%) and Cameroon (20.5%). However, higher incidences were reported in South Africa (31.7%), Tanzania (27.5%) and Southern Ethiopia (26.2%). Conversely, much lower incidences have been reported in Rwanda (3.2%), Ghana (4.3%) and Gabon (10.2%).<sup>19</sup> Furthermore, varying prevalences have been reported within Nigeria, with earlier studies reporting much lower prevalences than the more recent ones. Azzez *et al* reported a pooled prevalence of 11% (0.5–38%), found highest in the Federal Capital



**Table 4** Behavioural and lifestyle characteristics of pregnant women by their GDM status in Ibadan, Southwest Nigeria

|                                                                                        |            | GDM           |              | Unadjusted relative risk |               | Adjusted Relative Risk |               |
|----------------------------------------------------------------------------------------|------------|---------------|--------------|--------------------------|---------------|------------------------|---------------|
|                                                                                        | Total      | Present N (%) | Absent N (%) | 95% CI                   | P value       | 95% CI                 | P value       |
| Cumulative incidence                                                                   | 714        | 148 (20.7)    | 566 (79.3)   |                          |               |                        |               |
| Maternal body mass index                                                               |            |               |              |                          |               |                        |               |
| Underweight                                                                            | 30 (4.3)   | 3 (2.1)       | 27 (4.9)     | 0.56 (0.19 to 1.66)      | 0.29          | 0.66 (0.22 to 1.97)    | 0.457         |
| Normal weight                                                                          | 349 (50.4) | 63 (43.8)     | 286 (52.1)   | 1                        |               | 1                      |               |
| Overweight                                                                             | 193 (27.8) | 39 (27.1)     | 154 (28.1)   | 1.12 (0.78 to 1.60)      | 0.54          | 1.17 (0.79 to 1.73)    | 0.422         |
| Obese                                                                                  | 121 (17.5) | 39 (27.1)     | 82 (14.9)    | 1.79 (1.27 to 2.51)      | <b>0.001*</b> | 1.85 (1.26 to 2.30)    | <b>0.002*</b> |
| Alcohol consumption                                                                    |            |               |              |                          |               |                        |               |
| No                                                                                     | 613 (85.8) | 123 (83.1)    | 490 (86.6)   | 1                        |               |                        |               |
| Yes                                                                                    | 101 (14.2) | 25 (16.9)     | 76 (13.4)    | 1.23 (0.85 to 1.79)      | 0.272         |                        |               |
| Tobacco exposure                                                                       |            |               |              |                          |               |                        |               |
| No                                                                                     | 687 (96.2) | 140 (94.6)    | 547 (96.6)   | 1                        |               |                        |               |
| Yes                                                                                    | 27 (3.4)   | 8 (5.4)       | 19 (3.4)     | 1.45 (0.80 to 1.48)      | 0.221         |                        |               |
| Physical activity                                                                      |            |               |              |                          |               |                        |               |
| High                                                                                   | 350 (49.0) | 70 (47.3)     | 280 (49.5)   | 1                        |               |                        |               |
| Low                                                                                    | 364 (51.1) | 78 (52.7)     | 286 (50.5)   | 1.07 (0.80 to 1.48)      | 0.638         |                        |               |
| The mean duration of moderate-intensity activity per week (minutes)                    |            |               |              |                          |               |                        |               |
|                                                                                        | 25.3±22.7  | 24.9±23.8     | 25.3±22.4    | 0.99 (0.99 to 1.01)      | 0.845         |                        |               |
| Time spent viewing television during daytime (hours)                                   |            |               |              |                          |               |                        |               |
|                                                                                        | 2.5±3.5    | 2.9±3.9       | 2.4±3.3      | 1.02 (0.98 to 1.06)      | 0.179         |                        |               |
| Fast food habits                                                                       |            |               |              |                          |               |                        |               |
| Frequent                                                                               | 114 (16.3) | 25 (17.1)     | 89 (16.0)    | 1                        |               |                        |               |
| Infrequent                                                                             | 587 (83.7) | 121 (82.9)    | 466 (84.0)   | 0.93 (0.64 to 1.33)      | 0.750         |                        |               |
| Sleep duration per day (hours)                                                         |            |               |              |                          |               |                        |               |
|                                                                                        | 7.9±1.7    | 7.8±1.7       | 7.9±1.7      | 0.96 (0.89 to 1.04)      | 0.373         |                        |               |
| Bold values indicate a statistically significant association between exposures and GDM |            |               |              |                          |               |                        |               |
| GDM, gestational diabetes mellitus.                                                    |            |               |              |                          |               |                        |               |

Bold values indicate a statistically significant association between exposures and GDM  
GDM, gestational diabetes mellitus.

Territory (16.5%).<sup>20</sup> The high incidence in our study could be attributed to several factors, including the rise in the burden of obesity and diabetes in the population, higher detection rates due to increased awareness and screening and differences in screening methods.<sup>17 40</sup> IADPSG diagnostic criteria used in this study have a higher sensitivity, due to their low threshold values, leading to a higher incidence.<sup>40 42</sup>

The risk factors associated with GDM, especially in LMIC, are multifactorial. In this study, we examined sociodemographic, clinical, obstetric and behavioural factors among our study participants. Importantly, advanced maternal age, maternal obesity and a history of congenital anomaly were significant risk factors of GDM. However, other known risk factors such as a history of GDM and a family history of diabetes mellitus and a history of macrosomia were not statistically significant in this study. Advanced maternal age ( $\geq 35$  years) was associated with a significantly increased risk of GDM (aRR)

compared with younger women. The association between advanced maternal age and GDM is established in the literature.<sup>17 19 24</sup> Increasing age at marriage and conception has been observed among women of reproductive age in LMIC, especially in urban areas, and is associated with increased years of schooling.<sup>48 49</sup> The plausible reasons for the association between advanced maternal age and GDM include the changes in human physiology that occur with increasing age, such as a decline in pancreatic function and reduced insulin sensitivity, as well as the rise in obese individuals and other comorbidities with age.<sup>50</sup> Therefore, apart from promoting early marriage and conception among women of reproductive age, women of advanced age require close monitoring during antenatal care, including early screening for GDM.

We found that maternal obesity increased the risk of GDM by twofold. Researchers have also confirmed this association.<sup>20 51–54</sup> GDM is caused by pancreatic cell dysfunction that results in inadequate insulin production



**Table 5** Association between gestational diabetes and pregnancy outcomes in Ibadan, Nigeria

|                             |         |                          | Model 1                    |               | Model 2                       |         |
|-----------------------------|---------|--------------------------|----------------------------|---------------|-------------------------------|---------|
| Pregnancy outcomes          | N/N     | Cumulative incidence (%) | Crude relative risk 95% CI | P value       | Adjusted relative risk 95% CI | P value |
| Caesarean section           |         |                          |                            |               |                               |         |
| No GDM                      | 152/497 | 30.9                     | 1.00                       |               | 1.00                          |         |
| GDM                         | 41/125  | 33.3                     | 1.08 (0.81 to 1.43)        | 0.598         | 0.94 (0.71 to 1.24)           | 0.663   |
| Emergency caesarean section |         |                          |                            |               |                               |         |
| No GDM                      | 87/497  | 17.5                     | 1.00                       |               | 1.00                          |         |
| GDM                         | 15/125  | 12.0                     | 0.69 (0.41 to 1.14)        | 0.148         | 0.59 (0.41 to 1.14)           | 0.063   |
| Elective caesarean section  |         |                          |                            |               |                               |         |
| No GDM                      | 66/497  | 13.3                     | 1.00                       |               | 1.00                          |         |
| GDM                         | 26/125  | 20.8                     | 1.57 (1.04 to 2.36)        | <b>0.032*</b> | 1.32 (0.86 to 2.03)           | 0.199   |
| Spontaneous vertex delivery |         |                          |                            |               |                               |         |
| No GDM                      | 313/497 | 63.0                     | 1.00                       |               | 1.00                          |         |
| GDM                         | 77/125  | 61.6                     | 0.98 (0.84 to 1.14)        | 0.778         | 1.08 (0.92 to 1.27)           | 0.316   |
| Induction of labour         |         |                          |                            |               |                               |         |
| No GDM                      | 21/497  | 4.2                      | 1.00                       |               | 1.00                          |         |
| GDM                         | 6/125   | 4.8                      | 1.14 (0.47 to 2.75)        | 0.778         | 1.14 (0.43 to 2.99)           | 0.786   |
| Macrosomia                  |         |                          |                            |               |                               |         |
| No GDM                      | 22/448  | 4.9                      | 1.00                       |               | 1.00                          |         |
| GDM                         | 8/111   | 7.2                      | 1.47 (0.67 to 3.20)        | 0.337         | 1.25 (0.51 to 3.06)           | 0.620   |
| Low birth weight            |         |                          |                            |               |                               |         |
| No GDM                      | 33/448  | 7.4                      | 1.00                       |               | 1.00                          |         |
| GDM                         | 8/111   | 7.2                      | 0.98 (0.47 to 2.06)        | 0.954         | 0.64 (0.23 to 1.82)           | 0.412   |
| Preterm delivery            |         |                          |                            |               |                               |         |
| No GDM                      | 55/471  | 11.7                     | 1.00                       |               | 1.00                          |         |
| GDM                         | 17/111  | 14.3                     | 1.22 (0.74 to 2.02)        | 0.437         | 1.47 (0.81 to 2.67)           | 0.198   |
| Birth Asphyxia at 1 min     |         |                          |                            |               |                               |         |
| No GDM                      | 55/325  | 16.9                     | 1.00                       |               | 1.00                          |         |
| GDM                         | 15/86   | 17.4                     | 1.03 (0.61 to 1.73)        | 0.909         | 0.94 (0.53 to 1.67)           | 0.827   |
| Postpartum haemorrhage      |         |                          |                            |               |                               |         |
| No GDM                      | 79/497  | 15.9                     | 1.00                       |               | 1.00                          |         |
| GDM                         | 21/125  | 16.8                     | 1.06 (0.68 to 1.64)        | 0.806         | 0.82 (0.48 to 1.38)           | 0.462   |

n/N (number of events in the group)/total number of people in the group.

Model 1: association of GDM and pregnancy outcomes (unadjusted).

Model 2: association of GDM and pregnancy outcomes adjusted for maternal age, income and maternal BMI (adjusted).

\* Bold values indicate a statistically significant association between exposures and GDM

BMI, body mass index; GDM, gestational diabetes mellitus.

and the consequent inability to counter the insulin resistance from rising placental hormones. Additionally, maternal obesity worsens the risk of GDM by exaggerating the insulin-resistant state of normal pregnancy. Therefore, the exaggerated insulin resistance further stresses the pancreatic cells, resulting in a more severe GDM.<sup>55–57</sup> Fortunately, maternal obesity is a modifiable risk factor for GDM; hence, particular attention ought to be paid to women with obesity during pregnancy by providing

targeted antenatal and postnatal interventions, which include early screening for identifying GDM lifestyle modifications, such as nutritional counselling, encouraging physical activity and the prevention of excessive gestational weight gain, which are essential in preventing and managing GDM.<sup>54</sup>

Importantly, we found the women with a history of congenital anomaly had roughly a threefold (aRR) higher risk of GDM than women without that history. Congenital

anomaly is a structural, behavioural, functional or metabolic disorder that occurs during intrauterine life and is a major contributor to neonatal mortality in LMIC. The link between maternal hyperglycaemia and an elevated risk of congenital anomaly among offspring has been well reported in the literature.<sup>58–60</sup> However, the association between a history of congenital anomaly and GDM may also suggest an undetected prepregnancy metabolic dysfunction characterised by hyperglycaemia, insulin resistance and inflammation in the mother. An intra-uterine hyperglycaemic environment, especially during the embryonic stage, can increase the risk of congenital anomalies. Wu *et al*<sup>61</sup> reported a greater risk of congenital anomalies among women with prepregnancy hyperglycaemia in the USA.<sup>61</sup> Hence, preconceptual care, including risk assessment, screening and counselling, is recommended for women at risk for GDM during antenatal care.

We examined the association between GDM and selected pregnancy outcomes among our study participants. Studies that have examined the adverse pregnancy outcomes associated with GDM are uncommon in LMIC, such as Nigeria.<sup>62–64</sup> Remarkably, the incidences of CS, elective CS, induction of labour, macrosomia and preterm delivery were higher among women with GDM compared with women without GDM; none of these had significant associations with GDM. A study in the Netherlands (HIC) had also reported the lack of associations between GDM and adverse pregnancy outcomes.<sup>65</sup> Although this association may also be due to bias resulting from losses to follow-up. The finding may be associated with the quality of care provided at our study sites, which are comprehensive obstetric care facilities that had to manage GDM appropriately, potentially averting adverse pregnancy outcomes. The component of the GDM care includes glycaemic monitoring and control, restricting excessive weight gain through dietary modification and exercise, use of medications and planned interventional delivery (elective CS). The American College of Obstetricians and Gynaecologists recommends insulin as the standard pharmacological treatment for GDM.<sup>66</sup>

Moreover, the risk of an elective CS was higher among women with GDM (uRR=1.57) compared with women without GDM. GDM increases the risk of caesarean delivery through pregnancy complications, which are cephalo-pelvic disproportion, macrosomia, obstructed labour and perineal tear. Additionally, the risk of elective caesarean delivery is heightened among women with GDM because it is performed to avert adverse fetal outcomes such as unexplained intrauterine fetal death or stillbirth in the presence of poor glycaemic control and a large-sized fetus. Therefore, a planned preterm delivery after 34 weeks through elective CS is performed.<sup>54 67</sup> However, the association became insignificant after adjusting for confounders (aRR=1.32). Because GDM increases the risk of pregnancy complications, the goal of GDM management is to optimise the glycaemic control, ensure safe delivery and reduce perinatal complications. The models

of GDM care vary according to the levels of development of the healthcare system. However, several LMICs, such as Nigeria, adopt a more pragmatic and multifaceted approach to GDM management. This includes risk factor-based screening, self-monitoring of blood glucose and dietary and lifestyle modifications that aim for specific glycaemic targets. When these targets are not achieved, pharmacological treatment, primarily insulin, is required.<sup>68</sup>

The main strength of our study is the multicentre, prospective cohort study design, which facilitates the investigation of multiple risk factors and outcomes of GDM, reduces temporal bias and enhances generalisability to women who obtain care from comprehensive facilities in LMIC. The limitations included losses to follow-up and a low uptake of OGTT. The cohort of women in this study was urban-dwelling, highly educated and obtained care from high-level healthcare facilities; Hence, it may not be representative of women in the general population, especially those in rural areas or who obtain maternal care from primary healthcare centres. These biases are possible due to a potentially biased population, losses to follow-up and missing data from some pregnancy outcomes. Considerations for future research include screening for GDM in early pregnancy and examining treatment models of GDM in LMICs.

### Conclusion and implications of the study

GDM was widespread within our population, occurring in one in every five pregnant women obtaining care at comprehensive obstetric health facilities in LMICs. The significant risk factors of GDM in this study were advanced maternal age, maternal obesity and a history of congenital anomaly. These factors should be targeted for public health interventions, including encouraging early marriage and childbirth among women of reproductive age, lifestyle modification among pregnant women with obesity and early screening and diagnosis of GDM. Additionally, attention should be given to the prevention and management of GDM among women at high risk for GDM. Future interventions on GDM in Nigeria should support universal screening of GDM and improve access to prevention and control across all levels of the health system.

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**Ethics approval** This study involves human participants and was approved by We obtained the ethical approval for the study from the UCH/UI ethical review board (UI/EC/15/0060) and the Oyo State Ministry of Health Ethical Committee (AD/13/479/710. Permission to conduct the study was obtained from the head of the department and hospital unit. We got verbal and written informed consent from individual respondents before the commencement of the research. The study protocol and conduct adhered to the principles in the Declaration of Helsinki. Participants gave informed consent to participate in the study before taking part.

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